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E-260 CIRCULATING WATER SYSTEM CATHODIC PROTECTION

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of circulating water system cathodic protection in accordance with Part 4, Subsection 1.2 Summary of Work.

The Contractor shall provide and install complete Cathodic Protection System of Impressed Current type and/or Sacrificial Anodes type, in conjunction with the appropriate plant material selection, a continuous protection from corrosive action by the chlorine injected seawater, for the plant listed below. Interaction with surrounding plant grounding or other structures, shall be taken into account in the system design:

(Impressed current type)

- (1) Travelling Screen and Bar Screens and Trash Rakes.
- (2) Circulating Water Pump and Seawater Pump Casings, and Seawater Pipings.
- (3) Condenser Tube Cleaning System
- (4) All other corrosible portions of Seawater System, pipeworks and valves except the below.

(Sacrificial anode type)

The Contractor shall provide and install complete Sacrificial Anodes type Cathodic Protection System for the following equipment:

- (1) Condenser Water Boxes
- (2) Debris filter
- (3) Miscellaneous heat exchangers

Cathodic protection shall not cause damage to titanium structures by over protection. Further requirements for cathodic protection of equipment by means of the sacrificial anode method, such as steel sheet piling in direct contact with seawater are covered in Technical Specification E-261.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards:

2.1 National

PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation Indonesia).

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 British Standard

BS 7361 : Cathodic protection land and marine applications.

2.3 National Association of Corrosion Engineers

NACE RP0169 : Standard Recommended Practice for Contract of External Corrosion on Underground or Submerged Metallic Piping System.

2.4 National Electrical Manufacturers Association

NEMA MR-20 : Semiconductor Rectifiers, Cathodic Protection Units.

2.5 International Electrotechnical Commission

IEC 60076 : Power Transformers

IEC 60146 : Semiconductor Convertors.

IEC 60529 : Degrees of protection provided by enclosures (IP Code).

3. OPERATING AND DESIGN CONDITIONS

The Circulating water system cathodic protection systems shall prevent the corrosion of the associated equipment due to immersion in the seawater used in the circulating water system and seawater system.

The circulating water system cathodic protection systems shall be designed for outdoor operation as a part of combined cycle power plant located in tropical climate along the seacoast. Refer to Part 4, Subsection 1.1, 2.14 and 2.16 for

meteorological, seismic and environmental condition in the site.

The cathodic protection system shall be designed for a minimum of thirty (30) years operation, and shall include adequate margin for 20% coating deterioration, changes in soil/water characteristics.

4. DETAILED REQUIREMENTS

4.1 General

Power supply for cathodic protection systems will be supplied from Power Distribution Panel with 50 Hz, 400 V, 3 phase .

The cathodic protection system shall have adequate current output to prevent corrosion over the entire surface of the protected equipment. The cathodic protection systems shall be able to maintain a structure to sea water potential of not less than - 0.8 volt as per BS 7361 or equivalent with respect to a silver-silver chloride half cell.

The anode voltage shall be limited to not more than eight (8) volts to prevent deterioration of the titanium substrate.

The Impressed Current cathodic protection system shall consist of, but not be limited to, the following:

- (1) Transformer/rectifier unit including control equipment
- (2) Anodes
- (3) Reference cells
- (4) Cables, conduits and wiring
- (5) Approved quantity of cathodic protection test points

4.2 Transformer/Rectifier Unit

All Transformer/rectifier units shall be equipped with automatic potential control to maintain a constant protected object to seawater potential.

The control equipment shall be so arranged that each group of equipment shall be controlled independently of other equipment.

The transformer shall be three-phase, two-winding and self cooled. Dry type

transformers and shall have class B (or better) insulation to operate within the temperature rise limits for class B insulation.

The rectifier shall be equipped with step less voltage control, with coarse and fine adjustment, The voltage control shall also be manually operated.

The rectifier shall be a three-phase, full-wave bridge type, suitable for operation for 400 volt, 50 hertz, 3 phase supply.

The transformer/rectifier shall be equipped with controls to automatically regulate the metal-to-sea water potential to within ± 10 mV of its preset voltage to ensure adequate, but not excessive, protective current under all operating and seawater conditions. The control circuit shall be designed to minimize the error in the reference cell reading caused by the protection current. Connections to the reference cell shall be made by low current leads separate from the main power leads. Both manual and automatic voltage control shall be provided. The rectifier control circuit shall include a current limiter which shall protect the rectifier against short circuits or open circuits in the rectifier output circuit.

The transformer/rectifier units shall be equipped with an efficiency filter which shall limit the output ripple to 5% rms of the DC output, or less. In no case shall the ripple exceed a value which is not consistent with long life of platinized anodes.

The transformer/rectifier diodes and silicon controlled rectifiers shall be equipped with fuses and surge suppressors. Surge suppression shall be provided at the rectifier output.

The transformer/rectifiers units shall be equipped with two (2) electrically isolated single-pole double-throw contacts to indicate low, output or rectifier failure. The contact shall be suitable for operating in the 220 volt DC control circuits.

Each transformer/rectifier shall be equipped with a DC voltmeter and ammeter for monitoring the output of the rectifier. The meters of outdoor rectifiers shall be mounted in weather-proof enclosures equipped with a window to allow reading the meters without opening the enclosure. A convenient means of measuring the structure-to-sea water potential shall be provided.

Each transformer/rectifier units shall have filters or other means to prevent voltage or other distortion being fed back into the AC power supply due to the firing of the silicon controlled rectifiers or for any other reason; The filter shall limit the transient and harmonic voltage and currents to not more than 10% of the fundamental.

Each transformer/rectifier units shall be equipped with a three-pole AC circuit breaker at the AC line entrance.

A suitable over-temperature device shall be provided and which will trip the AC input circuit breaker of the rectifier cabinet if the temperature inside the enclosure reaches an unsafe value.

Cable terminals and lugs shall meet the requirement of Part 4, Technical Specifications E-001 and I-002.

There shall be one (1) DC positive terminal and one (1) DC negative terminal, suitable for single conductor insulated cable. One (1) reference cell terminal for each reference cell used in the system and one (1) structure potential terminal shall also be provided for 1.5 mm² cable. A switch shall be provided to switch between reference cells.

All transformer/rectifier enclosure internal wiring shall meet the requirements of Part 4, Technical Specification E-280.

Each transformer/rectifier enclosure shall be equipped with an outside nameplate in accordance with Part 4, Technical Specification E-280.

Each transformer/rectifier enclosure shall be equipped with the manufacturer's standard nameplate, which shall contain the following information as a minimum:

- (1) Manufacturer's name
- (2) Model, type, and serial number
- (3) Alternating current voltage and frequency
- (4) Number of phases

- (5) Alternating-current ampere rating
- (6) Direct-current ampere rating (maximum and minimum)
- (7) Direct-current ampere rating
- (8) Type, manufacturer and model number of reference electrode(s).

The nameplate may be mounted internally for indoor rectifiers, but shall be mounted on the exterior of outdoor rectifiers.

A removable cover shall be provided for access to equipment, if necessary, for replacement or adjustment. The front door shall furnish access to circuit breaker handles, instruments, rectifier DC terminals, reference voltage measurement terminals, reference cell toggle switch, manual and automatic control adjustments.

All live parts of 110 volts or over shall be protected from contact with persons opening the front door.

An operating instruction manual, including a wiring diagram of the rectifier equipment, shall be on the inside of the front door or at another convenient location.

Each transformer/rectifier units cabinet shall be equipped with a NEMA two hole ground pad (for drilling pattern).

Transformer/rectifier units enclosures shall be free standing and self supporting. The degree of protection shall be IPW 55 or better according to IEC 60529.

The rectifier enclosures shall be equipped with windows so that it is possible to read monitoring voltmeters and ammeters without opening the doors. The enclosures shall be paint finished suitable for marine service. Each transformer/rectifier unit shall be equipped with all necessary reference electrodes required for operation.

An appropriate cooling system shall be provided for transformer/rectifier unit.

4.3 Anodes

All anodes shall be platinized titanium.

The platinum coating shall be applied to form a solid, void-free film over the substrate and shall have adequate thickness for thirty (30) years of operation. The minimum platinum thickness shall be 5 microns.

The anodes shall be completely assembled, complete with all accessories. All anodes for protecting the Circulating Water System shall be of flange type coupling in order not to damage the lining.

4.4 Reference Cells

All reference cells shall be of zinc.

Reference cells for use inside pipes shall be mounted on couplings identical to those used for anodes. The reference cells shall have adequate surface area to prevent polarization.

4.5 Cables and Wiring

Cabling and wiring installation and the use of conduit, junction boxes, flexible connection, etc. shall be as required by Part 4, Technical Specification E-001 and E-232.

4.6 Cathodic Protection Test Points

To measure the current and potential of protected plants, the Contractor shall provide Cathodic Protection Test Points, in addition to test facilities in transformer/rectifier units.

Cathodic Protection Test Point shall be suitable for current and voltage measurements using portable measuring equipment and reference cells. Enclosures shall be IPW 55 according to IEC 60529.

4.7 Tools and Test Equipment

The Contractor shall provide one (1) set of all special tools needed to maintain the cathodic protection system, including the rectifiers. These tools shall be new and shall not be used by the Contractor for the erection and commissioning of the Works. The tools provided shall include a potentiometric voltmeter and two (2) silver-silver- chloride half cells for conducting corrosion surveys.

4.8 Calculations

All design calculations shall be submitted to the Owner for review and approval.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 Shop Tests

Shop tests of the electrical equipment for cathodic protection shall be in accordance with the relevant standard of each equipment.

5.2 Site Tests and Corrosion Surveys

After installation, the Contractor shall perform commissioning tests to ensure that the equipment is functioning correctly. The tests shall include measurements of anode currents, rectifier voltage, and protected equipment voltages with respect to a reference half cell.

The Contractor shall perform corrosion surveys for the system to inspect and adjust the cathodic protection system at the time of six (6) months and one (1) year after commissioning to ensure the system continuously to provide effective corrosion protection.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for the circulating water system cathodic protection shall conform to the requirements of Part 4, Subsection 2.7. The procedures as specified, shall include, but not be limited to, the followings:

- (1) Packaging and shipping
- (2) Shop test procedures
- (3) Cleaning and painting

The documentations as specified shall include, but not be limited to, the followings:

- (1) Shop test reports
- (2) All other documentation required by applicable codes and standards.

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